25900550

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) Regular EXAMINATION October 2025

Third SEMESTER

Discipline Specific Core Course (DSC) - MG3DSCPES201 –

UNDERSTANDING HUMAN BODY

(2024 ADMISSION ONWARDS)

Answer key

Duration: 1 Hour 30 Minutes Maximum Marks: 50

Remember(K), Understand(U), Apply(A), Analyse (An), Evaluate(E), Create(C), Skill(S), Interest (I) and Appreciation (Ap)

Students s hould attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

MCQ

Answer all questions

Each question carries 1 mark

1. Mitochondria are abundant in cells that perform **[U]/[CO1]**
d). Energy production

2. Which bone articulates with the glenoid cavity to form the shoulder joint? **[U]/[CO2]**
d). Humerus

3. Which molecule provides the energy for muscle contraction? **[A]/[CO1]**

b). ATP

4. Which type of neuron is found in the spinal cord and connects sensory and motor neurons?

d). Interneuron **[U]/[CO1]**

5. What is the basic structural and functional unit of the nervous system? **[U]/[CO1]**

b). Neuron

6. Which part of the brain controls the rate of breathing? **[U]/[CO2]**

b). Medulla oblongata

7. Which term refers to moving a limb away from the midline of the body? **[An]/[CO2]**

c). Abduction

8. Which muscle fiber has the highest mitochondrial content? **[K]/[CO1]**

a). Type I

9. In double circulation, blood passes through the heart **[U]/[CO2]**

b. Twice

10. The nervous system is broadly divided into how many main parts? **[U]/ [CO1]**

b). Two **[1x10= 10]**

Part B

SHORT ANSWER QUESTIONS

Answer any 10 questions

Each question carries 2 marks

11. What is blood classified as? **[U]/[CO1]**

Ans: Blood is classified as a connective tissue because it connects different parts of the body by transporting oxygen, nutrients, hormones, and waste products. It consists of two main components: plasma (the liquid part) and formed elements (red blood cells, white blood cells, and platelets). Although it is a fluid, it performs the same functions as other connective tissues—binding and supporting body structures.

12. What is the function of the spinous process? **[An] / [CO6]**

Ans: The function of the spinous process is to:

Provide attachment points for muscles and ligaments of the spine, which help in movement and maintaining posture.

Act as a lever for muscles that move and stabilize the vertebral column.

13. Define muscle fiber fatigue. **[U]/[CO1]**

Ans: Muscle fiber fatigue is defined as the decline in a muscle's ability to generate force or power after prolonged or repeated activity.

It occurs due to factors like: Lack of energy (ATP depletion), Accumulation of lactic acid, Reduced oxygen supply

14. What is the role of the pituitary gland? **[U]/ [CO2]**

Ans: The pituitary gland is known as the "master gland" of the body because it controls the activities of many other endocrine glands.

Role of the pituitary gland: It secretes hormones that regulate growth, metabolism, and reproduction.

It controls other glands such as the thyroid, adrenal glands, and gonads (ovaries and testes).

15. How does the urinary system remove nitrogenous wastes? **[A]/ [CO2]**

Ans: The urinary system removes nitrogenous wastes like urea, uric acid, and creatinine from the body through a process of filtration, reabsorption, and excretion.

Process:

1. The kidneys filter blood to remove nitrogenous wastes and excess water.

2. These wastes form urine.

3. Urine then passes through the ureters to the urinary bladder, where it is stored.

4. Finally, it is excreted through the urethra.

16. Give two examples of anatomical directional terms. **[U]/[CO5]**

Ans: 1. Superior (Cranial) – Above or higher in position.

2. Inferior (Caudal) – Below or lower in position.

3. Anterior (Ventral) – Toward the front of the body.

4. Posterior (Dorsal) – Toward the back of the body.

5. Medial – Toward the midline of the body.

6. Lateral – Away from the midline of the body.

7. Proximal – Closer to the point of attachment or origin.

8. Distal – Farther from the point of attachment or origin.

17. Give two general features of the scapula. **[U]/ [CO2]**

Ans: 1. The scapula is a large, flat, triangular bone located on the posterior side of the thorax between the 2nd and 7th ribs.

2. It has two main surfaces (costal and dorsal), three borders (Superior border, Medial border, Lateral border), three angles (Superior angle, Inferior angle, Lateral angle) and three processes (spine, acromion, and coracoid).

18. State the function of actin and myosin in muscle contraction. **[A] / [CO2]**

Ans: Actin: Forms thin filaments that provide sites for myosin to attach during contraction.

Myosin: Forms thick filaments with heads that bind to actin and pull the filaments inward, causing the muscle to shorten and contract.

19. Give two examples of mixed nerves. **[U]/[CO1]**

Ans: 1. Spinal nerves – All 31 pairs of spinal nerves are mixed nerves.

2. Trigeminal nerve (Cranial nerve V) – Carries both sensory and motor Fibers.

3. Facial nerve (Cranial nerve VII) – Controls facial muscles (motor) and taste sensations (sensory).

4. Glossopharyngeal nerve (Cranial nerve IX) – Involved in taste (sensory) and swallowing (motor).

5. Vagus nerve (Cranial nerve X) – Supplies motor and sensory fibers to many organs like the heart and lungs.

20. What is peristalsis? **[U]/[CO2]**

Ans: Peristalsis is the wave-like muscular movement of the walls of hollow organs such as the esophagus, stomach, and intestines that pushes food and liquids forward through the digestive tract.

21. Define the term 'caudal'. **[U]/[CO5]**

Ans: Caudal is an anatomical directional term that refers to a position toward the tail end or the inferior (lower) part of the body.

The spinal cord is caudal to the brain.

22. What is a ball and socket joint? Give an example. **[U]/ [CO2]**

Ans: A ball and socket joint is a type of synovial joint in which the rounded head (ball) of one bone fits into the cup-like cavity (socket) of another bone, allowing movement in all directions, including rotation.

Shoulder joint (between humerus and scapula)

Hip joint (between femur and hip bone)

[2x10 = 20]

Part C

SHORT ESSAY QUESTION

Answer any 4 questions

Each question carries 5 marks

23. Discuss the functional classification of muscles with examples. **[U]/[CO1]**

Ans: Muscles can be classified functionally based on the type of movement they produce during an action. There are four main functional types:

1. Agonist (Prime Mover): The main muscle responsible for producing a specific movement. It contracts to cause the desired action.

Example: The biceps brachii acts as the agonist during elbow flexion.

2. Antagonist: The muscle that opposes the action of the agonist. It relaxes or lengthens when the agonist contracts.

Example: The triceps brachii is the antagonist during elbow flexion (it opposes the biceps).

3. Synergist: Muscles that assist the agonist in performing a movement. They help by adding extra force or reducing unnecessary movement.

Example: The brachialis and brachioradialis assist the biceps brachii in elbow flexion.

4. Fixator (Stabilizer): Muscles that stabilize the origin of the agonist so that it can act more efficiently.

Example: The trapezius and rhomboid muscles stabilize the scapula while the deltoid moves the arm.

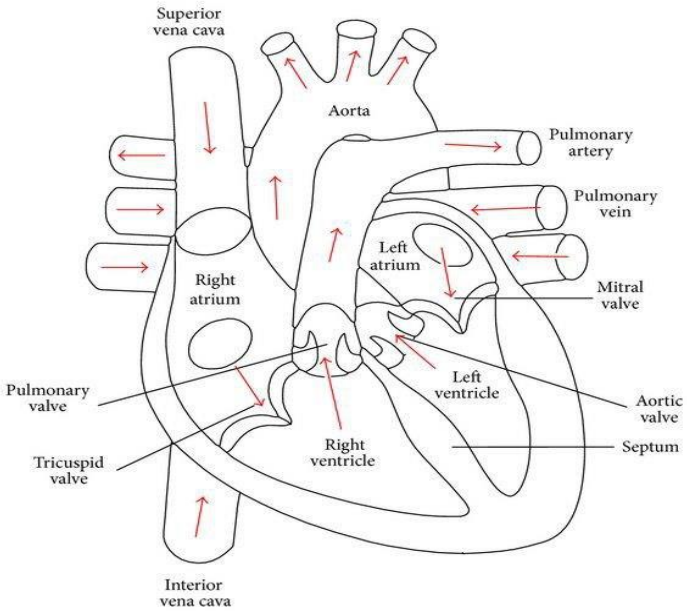
24. Explain the structure and functions of the heart with a neat diagram. [U]/[CO2]

Ans: Structure:

- The heart is a muscular, cone-shaped organ located in the thoracic cavity, between the lungs.
- It is made of cardiac muscle and is divided into four chambers:
- Right atrium and right ventricle (receive and pump deoxygenated blood)
- Left atrium and left ventricle (receive and pump oxygenated blood)
- Valves (tricuspid, pulmonary, mitral, and aortic) ensure one-way flow of blood.
- The septum divides the heart into right and left sides.
- The heart is enclosed in a protective sac called the pericardium.

Functions:

- Pumping Blood: The heart acts as a pump to circulate blood throughout the body.
- Oxygen Transport: It sends deoxygenated blood to the lungs for oxygenation and oxygenated blood to the body.
- Maintains Blood Pressure: Regular contractions maintain proper blood pressure and flow.
- Removes Waste: It helps transport carbon dioxide and other waste products for elimination.



25. Explain the role of the spinal cord in reflex action and communication. [U]/[CO1]

Ans: 1. In Reflex Action:

- The spinal cord acts as the center for reflexes.
- During a reflex action (like pulling your hand away from a hot object), the sensory neurons carry impulses from receptors to the spinal cord.
- The spinal cord quickly processes this information through interneurons and sends a response through motor neurons to the muscles — without involving the brain.
- This allows the body to respond rapidly to danger or stimuli.

- Example: Touching something hot → sensory impulse to spinal cord → motor impulse to hand muscles → hand withdraws.

2. In Communication:

- The spinal cord serves as a communication pathway between the brain and the rest of the body. It transmits sensory information from body parts to the brain and motor commands from the brain to muscles and glands.
- This ensures coordination and control of voluntary and involuntary body functions.

26. Discuss the structural features and functions of neutrophils. [S]/[CO3]

Ans: 1. Structural Features:

- Type: Neutrophils are a type of white blood cell (WBC), specifically granulocytes.
- Nucleus: They have a multi-lobed nucleus (usually 2–5 lobes), which gives them the name polymorphonuclear leukocytes.
- Cytoplasm: The cytoplasm contains fine granules that stain lightly with both acidic and basic dyes (hence the name neutrophil).
- Size: They are about 10–12 micrometers in diameter.
- Mobility: Neutrophils are highly motile and can move through capillary walls into tissues by diapedesis.

2. Functions:

- First line of defence: Neutrophils are the first immune cells to reach the site of infection or injury.
- Phagocytosis: They engulf and destroy bacteria and dead cells, helping to clear infection.
- Enzyme release: Their granules contain digestive enzymes (like lysozyme and proteases) that help break down pathogens.
- Inflammatory response: They release chemicals that attract other immune cells and promote inflammation.
- Short lifespan: Neutrophils live for only a few hours to a few days, sacrificing themselves during infection.

27. Explain bone remodeling and factors that influence it [U]/[CO2]

Ans: Bone remodeling is the continuous process of breaking down old bone tissue and forming new bone tissue to maintain bone strength and mineral balance. It helps in repairing micro-damages, reshaping bones after stress, and regulating calcium levels in the blood.

Steps in Bone Remodeling:

- Resorption: Osteoclasts break down and remove old or damaged bone tissue.
- Reversal: The bone surface is prepared for new bone formation.
- Formation: Osteoblasts build new bone matrix and deposit minerals.
- Resting phase: The new bone becomes mature and remains until the next remodeling cycle.

Factors Influencing Bone Remodeling:

- Hormones – Parathyroid hormone (PTH), calcitonin, growth hormone, and sex hormones (estrogen, testosterone) regulate bone resorption and formation.
- Mechanical stress or exercise – Physical activity and weight-bearing exercise strengthen bones by stimulating bone formation.
- Nutrition – Adequate calcium, vitamin D, and protein are essential for bone health.

- Age – Remodeling slows with age, leading to bone loss.
- Genetics – Determines bone density and structure.
- Diseases and medications – Conditions like osteoporosis or drugs such as steroids can affect remodeling.

28. Explain irritability in muscles and its importance for survival. **[U]/[CO1]**

[5x4 = 20]

Ans: Irritability (also called excitability) is the ability of a muscle or nerve tissue to respond to a stimulus, usually a nerve impulse (such as an electrical or chemical signal). It is one of the essential properties of living tissues, especially in muscles and nerves.

When stimulated, the muscle generates an electrical signal that leads to contraction.

Explanation:

Muscles receive signals from the nervous system through motor neurons.

These signals (stimuli) cause chemical and electrical changes in the muscle fibers. This leads to muscle contraction, enabling movement or other vital functions (like heartbeat or breathing).

3. Importance for Survival:

Movement: Allows the body to move away from danger or perform daily activities.

Protection: Helps in quick reflex responses to harmful stimuli (e.g., pulling away from something hot).

Circulation: Cardiac muscle irritability ensures continuous heart contractions to pump blood.

Respiration: Skeletal muscles like the diaphragm contract rhythmically to maintain breathing.

Posture and Stability: Maintains body position and balance through constant, mild muscle contractions.